

DATA STRUCTURE

LAB PROGRAMS

22. Dijkstra's Algorithm

```
#include <stdio.h>

#define V 4

#define INF 9999

int main() {
    int graph[V][V] = {
        {0,4,0,0},
        {4,0,8,2},
        {0,8,0,5},
        {0,2,5,0}
    };

    int dist[V], visited[V]={0};

    for(int i=0;i<V;i++)
        dist[i]=INF;

    dist[0]=0;

    for(int count=0; count<V-1; count++) {
        int min=INF,u;

        for(int i=0;i<V;i++)
            if(!visited[i] && dist[i]<min) {
                min=dist[i];
                u=i;
            }

        visited[u]=1;

        for(int v=0;v<V;v++) {
            if(!visited[v] && graph[u][v] &&
```

```

        dist[u]+graph[u][v] < dist[v])
        dist[v]=dist[u]+graph[u][v];
    }
}
printf("Vertex Distance\n");
for(int i=0;i<V;i++)
    printf("%d\t%d\n",i,dist[i]);
return 0;
}

```

Output

```
Vertex Distance
```

```
0    0
```

```
1    4
```

```
2   11
```

```
3    6
```

```
=== Code Execution Successful ===
```